CS3010: Software Engineering Lab

Assignment 7 Report: Implementation of RSPC Module in Fusion ERP

1. Objective and Problem Statement

The objective of this assignment is to develop and integrate the Research, Sponsored Projects and Consultancy (RSPC) module into the Fusion ERP system. The module provides a streamlined interface for faculty members and administrators to handle sponsored research project submissions, consultancy project registrations, financial requisitions, and project progress reporting. In accordance with the modular architecture of Fusion ERP, the module operates as an independent domain module and avoids direct cross-module foreign key dependencies.

2. Implemented Features and Business Rules

The implementation includes the following key functional capabilities based on the module requirements:

? **Sponsored Research Proposals:** Allows faculty to create and submit project proposals including funding agency, budget, and abstract. Implemented a multi-tier approval workflow where proposals move from Submitted to HoD Endorsed and Dean Sanctioned.

? **Consultancy Projects:** Faculty can register consultancy assignments with client details. Implemented automatic calculation of the 30% institute overhead share and 70% investigator share in the model upon saving.

? **Financial and Procurement Requisitions:** Faculty can submit fund disbursement, procurement, and travel requests. The system automatically flags any request exceeding 5 Lakhs (500,000 INR) as requiring Director approval.

? **Project Progress Reports:** Facility to submit annual progress reports and project completion reports.

3. Technical Implementation

Backend (Django and Django REST Framework):

? Defined four core models in modules/rspc/models.py: ResearchProposal, ConsultancyProject, FinancialRequest, and ProjectReport.

? Created ModelSerializers in api/serializers.py to format and validate input data.

? Created ModelViewSets in api/views.py with custom approval actions (hod_approve, dean_approve) and user-scoped querysets.

? Configured URL routing in api/urls.py using DefaultRouter and connected the endpoints to config/urls.py under /api/rspc/.

? Registered the application in config/settings/base.py inside DOMAIN_MODULES.

Frontend (React and Mantine):

? Created the primary dashboard component in client/src/modules/rspc/pages/RSPCDashboard.tsx with dedicated tabs for Research Proposals, Consultancy Projects, Financial Requests, and Project Reports.

? Implemented modal dialogs for submitting proposals, adding consultancy projects with live revenue split preview, and raising financial requests.

? Added an Axios API client in client/src/modules/rspc/api.ts with authentication token interceptors.

? Registered the module route in client/src/app/router.tsx and registered the module metadata in client/src/app/registry.ts.

4. Testing and Verification

The implementation was tested to ensure proper functionality and integration:

? **Django Unit Tests:** Added test cases in modules/rspc/tests/test_views.py covering proposal creation and approval, the 30/70 consultancy revenue split calculation, and the 5 Lakhs financial threshold check. All 3 tests executed and passed successfully.

? **Django System Check:** Ran python manage.py check, resulting in 0 errors and 0 warnings.

? **Database Migrations:** Created and applied the initial migration (0001_initial.py) cleanly without migration conflicts.

? **Frontend Typecheck:** Ran npm run typecheck with TypeScript compiling cleanly with 0 errors.

? **Production Build:** Ran npm run build with Vite generating the production bundle successfully.